

- Program executables (file system commands)
- Linux aliases
- Linux shell functions
- Linux shell reserved words
- Built-in commands

Best practices for writing effective scripts

Shell scripting, though very simple, is a very effective method for creating easy, small but effective programs. However, to write an efficient format of code script, one needs to take care of certain things in order to ensure better administration, management and debugging of the code, if needed.

Here are some points to be pondered while

writing a shell script:

- Write comments in the code wherever necessary. This helps others also to understand the working mechanism of the code much more easily.
- Try to exit the script as soon as any error is encountered within it, as even the smallest error bug can do great damage to your system.
- Use the functions you can write by yourself in the script, as it is much easier to call it and debug it.
- Using the \$ sign is recommended, in place of single quotes (') as it helps us to differentiate between the process of substitution and the declaration of commands.

Command line tips and tricks

- **History command:** This helps us check the back-dated history of almost all the commands fired by the user from the start.
- **Multiple commands in a single row:** We can write multiple commands in a single row by entering the terminator operator ';' or the AND operator '&&' after each command and hence execute multiple lines at a time.
- **Read log:** Using the 'tail' command, we can read the logs of any file in real-time.
- **Read Zip files:** The 'z' command helps us to also read Zip files of all formats even without extracting the actual data.
- **Alias and Grep commands:** These are basic Linux/UNIX commands that can help read and edit files and syntaxes easily.

The advantages and disadvantages of shell scripting

The advantages

- A special feature is that the user can easily run multiple commands together in a single line command.
- The commands are easy to write and execute, and are used in many automated systems.
- Previously executed operations can also be rewound, and the functions and commands can be performed automatically.
- Supports almost every system that has a UNIX based architecture and hence is portable from one system to another.

The disadvantages

- The execution time is longer as it is not specifically a programming language but a command line script.
- A new thread or mini process is created for each command executed in the shell script.

Comparison with other forms

Shell is a UNIX based command line compiling interface and hence it is somewhat tough for users to remember all the commands, whereas the other languages have only syntaxes and therefore are a bit easier.

However, taking a broader view, the shell is more effective in performing almost all the tasks possible. 

References

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